

Asrita Bobba

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EDUCATION

New York University, Tandon School of Engineering

New York, NY

Master of Science in Computer Engineering

- **Relevant Coursework:** Machine Learning, Deep Learning, Real-Time Embedded Systems, Computing Systems Architecture, Neuroinformatics, MLOps & Systems Engineering, Natural Language Processing, Advanced Topics in Computer Vision, Reinforcement Learning & Optimal Control I & II

KL University

Vijayawada, India

Bachelor of Technology in Artificial Intelligence & Data Science

- **Relevant Coursework:** Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Speech Processing, Signal Processing, Big Data Analytics, Cloud Computing, Data Structures & Algorithms, Probability & Statistics for Data Science, Computer Networks & Security, Operating Systems, Automata Theory & Compiler Design, Object-Oriented Programming

EXPERIENCE

Research Assistant

Aug. 2025 – Present

SAVVY Lab, New York University

New York, NY

- Conduct quantitative analysis of multimodal neural and behavioral data — including fNIRS and fMRI signals — to investigate anticipatory and social-cognitive mechanisms underlying stuttering variability.
- Develop and maintain data processing pipelines for time-series neural signals; apply statistical modeling and feature extraction to high-dimensional, multi-subject datasets.
- Collaborate across an interdisciplinary team of speech-language scientists, and clinicians to design experiments and interpret findings.

Data Analyst Intern

Jul. 2023 – Jan. 2024

Let's Transport

Bengaluru, India

- Built and fine-tuned predictive models for route optimization using Python (Pandas, NumPy, Scikit-learn), reducing travel time variance and improving logistics efficiency across the fleet.
- Designed interactive dashboards in Power BI and Tableau to surface operational KPIs, enabling data-driven decision-making for the strategy team.
- Processed and analyzed large-scale trip and vehicle datasets via SQL and Python, identifying performance trends that directly informed quarterly business planning.

Web Development Intern

Jun. 2022 – Jul. 2022

Swecha

Vijayawada, India

- Developed web applications with Django, HTML, CSS, and JavaScript to expand digital learning access for underserved students across multiple institutions.
- Architected scalable backend services, optimized web performance, and contributed to code reviews in a cross-functional team environment.
- Collaborated with designers and educators to translate pedagogical requirements into functional, accessible product features.

Environmental Data Intern

Dec.2022 – Dec. 2022

Asian Institute of Thailand

Khlong Nueng, Thailand

- Conducted structured field surveys and environmental data collection to analyze plant density and biodiversity across campus ecosystems.
- Applied statistical modeling in R and Excel to evaluate population dynamics and environmental sustainability indicators from collected data.
- Synthesized findings into a written report communicating methodology, results, and recommendations to a mixed technical and non-technical audience.

PROJECTS

Inference-Time Reasoning in Instruction-Tuned LLMs | *NLP, FLAN-T5, HuggingFace, HPC*

- Designed and evaluated inference-time self-improvement methods (Zero-Shot, Few-Shot, Chain-of-Thought, SGE) on FLAN-T5, analyzing how prompting strategies interact with model scale across arithmetic, QA, and commonsense reasoning tasks.
- Built a modular experimental framework with prompt abstractions, ablation studies, and qualitative error analysis, surfacing fundamental limitations of inference-time reasoning in encoder-decoder architectures.

Cross-Domain Image Super-Resolution | *PyTorch, ESRGAN, Diffusion Models, HPC*

- Benchmarked CNN-, GAN- (ESRGAN), and diffusion-based SR models on non-natural imagery, comparing reconstruction fidelity via PSNR, SSIM, and perceptual metrics across multiple scientific domains.
- Built scalable evaluation framework with custom degradation pipelines and ablation studies; findings guided architecture selection for robust cross-domain reconstruction.

Taxi Demand Prediction MLOps Pipeline | *FastAPI, ONNX, Airflow, Docker, Prometheus*

- Developed an end-to-end MLOps pipeline predicting high-demand NYC taxi pickup zones; containerized model serving with FastAPI + Docker and automated batch inference via Airflow DAGs.
- Deployed ONNX-optimized models for latency benchmarking; integrated Prometheus/Grafana monitoring and retraining feedback loops for continuous evaluation.

Connectome-Guided Reservoir Computing | *Python, PyTorch, conn2res, HPC*

- Led design and evaluation of biologically inspired MSS/SNN/ESN reservoir architectures using human and *Drosophila* connectomes, achieving perfect classification accuracy on structured spike-train manifolds.
- Collaborated with NeuroAI researchers to explore cross-species computational principles; built full pipeline including criticality tuning, PCA/raster visualization, and functional-network ablation studies.

Real-Time Embedded Sentry System | *C, PlatformIO, IMU, Embedded ML*

- Engineered an embedded security system using IMU-based gesture recognition for secure access control, achieving >79% top-1 accuracy under robust training with adversarial vulnerability analysis.

RISC-V Cycle-Accurate Processor Simulator | *Python, Computer Architecture*

- Implemented single-stage and 5-stage pipelined 32-bit RISC-V processor simulator in Python; designed hazard resolution for RAW and control-flow dependencies and evaluated CPI and execution cycle metrics.

LEADERSHIP & EXTRACURRICULAR

Vice President

2024 – Present

ACM-W (Association for Computing Machinery – Women), NYU

- Organized technical workshops, interview prep sessions, and panel discussions; managed event logistics to increase student engagement across NYU's engineering community.
- Coordinated cross-departmental outreach initiatives to promote inclusivity and professional development for women in computing at NYU.

Project Lead

2021 – 2024

Department Student Body, KL University

- Spearheaded planning and execution of coding workshops, hackathons, and industry networking sessions for 200+ students.
- Facilitated sessions connecting students with industry professionals, fostering mentorship opportunities and career development.

TECHNICAL SKILLS

Languages: Python, C++, C, Java, SQL, MATLAB, JavaScript

ML / AI Frameworks: PyTorch, Scikit-learn, HuggingFace Transformers, NumPy, Pandas, conn2res

MLOps & Infrastructure: Docker, Apache Airflow, FastAPI, ONNX, Prometheus, Grafana, AWS/GCP (cloud coursework)

Web & Backend: React.js, Django, Flask, Vue.js, Node.js, HTML/CSS, Firebase, MySQL

Tools & Platforms: Git, GitHub/GitLab, Linux, PlatformIO, Power BI, Tableau, R

Soft Skills: Cross-cultural communication (studied across 12 schools in 4 countries), public speaking, agile project management, research & technical writing

Languages: English, Hindi, Telugu (all native)